Unreal Is Not Open: prediction, horizons and the future in relativistic becoming

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Abstract

Local theories of relativistic becoming make what is real from an event its causal past. Does the unreality of the future from an event make the future open from it? We separate the two questions. Call an event determined from b if every past-inextendible causal curve from it meets b's chronological past; under deterministic laws with the domain-of-dependence property, such events are fixed by what is real from b. The notion is essentially Geroch's domain of prediction. We characterize it by a condition on the slices of any Cauchy time function, show that it is conformally invariant, and compute it for the standard cosmological models. In Minkowski spacetime, spatially flat and open big-bang universes, the closed radiation universe and de Sitter spacetime, nothing outside an event's past is ever determined from it. In the closed dust model, nothing is while the universe expands and everything is from every event after maximum expansion, the known moment at which the particle horizon disappears. So unreality and openness come apart in both directions, and, given the laws, which combination holds is a matter of global causal structure and can change with cosmic epoch. A closing remark records what the corresponding standpoint logic does and does not show about cosmological horizons.

1 Introduction

Stein's theorem says that in Minkowski spacetime a transitive, symmetry-invariant relation of becoming that makes some past event definite is the causal order or trivial [6, 18, 19]. A companion paper [20] extends this to general relativity under a locality principle: what has become for an event b always includes its chronological past and never goes beyond its causal past. On views that read "has become for b" as "is real from b" [7, 13], the future of b is not real from b.

It is tempting to infer that the future is *open* from b. The inference fails, and the way it fails is instructive. Openness in the sense relevant here is a matter of what the real fixes, and that depends on the laws and on the global causal structure. The point is old in outline: growing-block theorists who are determinists hold that the future is unreal yet fixed [4], and Rakíc's sea-fight case turns on an event outside an observer's past that a decision inside it appears to settle [16]. What is missing is a precise account, for relativistic spacetimes, of when what is real from an event fixes anything beyond itself. Geroch's domain of prediction supplies the tool [8], and Manchak's work on prediction its limits [14]. We use them to answer three questions:

1. Exactly which events are fixed, under deterministic laws, by what is real from b (Theorem 3)?
2. In the standard cosmological models, how much is (Section 3)?
3. What follows for the relation between unreality and openness (Section 4)?

Conventions. Spacetimes are smooth, connected, time-oriented and globally hyperbolic. $I^-_{\mathrm{s}}(p)$ is the strict chronological past (via timelike curves), $J^-(p)$ the causal past; $x \le y$ means $x \in J^-(y)$, and $x \ll y$ means $x \in I^-_{\mathrm{s}}(y)$. We use that $\le$ is closed, that $J^-(x) \cap S$ is compact for every Cauchy surface S, push-up ($y \le z \ll b$ implies $y \ll b$), and that $M = I^-_{\mathrm{s}}(S) \cup S \cup I^+_{\mathrm{s}}(S)$ for a Cauchy surface S [15]. A *Cauchy time function* is a continuous $t : M \to (\alpha, \beta)$, $-\infty \le \alpha < \beta \le \infty$, strictly increasing along future-directed causal curves, whose level sets $S_c = t^{-1}(c)$ are Cauchy surfaces [5].

2 Determined from an event

Definition 1. An event x is *determined from* b if every past-inextendible causal curve from x meets $I^-_{\mathrm{s}}(b)$. Write $\mathrm{Det}(b)$ for the set of such events. Clearly $I^-_{\mathrm{s}}(b) \subseteq \mathrm{Det}(b)$.

Remark 2 (What the definition does and does not say)**.** $\mathrm{Det}(b)$ is a property of the spacetime and the event. It becomes a claim about what is fixed only given laws with the *domain-of-dependence property*: data on a region fix the fields at every event all of whose past-inextendible causal curves meet the region. Under such laws we call the events of $\mathrm{Det}(b)$ *nomologically settled from* b. This is not Belnap's notion of settledness, on which a proposition is settled at a point event if it holds in every possible history through it [2]; under indeterministic laws the two come apart, and $\mathrm{Det}(b)$ says nothing about them. The domain-of-dependence property is also a locality assumption on the dynamics. We use $I^-_{\mathrm{s}}(b)$, which every local becoming theory counts as real from b, rather than $J^-(b)$; the characterization below uses that $I^-_{\mathrm{s}}(b)$ is open.

Relation to Geroch. Geroch's domain of prediction $P(q)$ is the set of x such that every past-directed timelike curve from x without past endpoint enters $I^-(q)$, but $I^-(x) \not\subseteq I^-(q)$ [8, sec. 3]. $\mathrm{Det}(b)$ is the causal-curve analogue of the union of $P(b)$ with the events whose pasts lie in that of b; the two differ at most on null boundaries. Geroch gives an equivalent form by achronal surfaces in $I^-(q)$ [8, sec. 3]; Theorem 3 is a slicing form for globally hyperbolic spacetimes.

Theorem 3. *Let $t : M \to (\alpha, \beta)$ be a Cauchy time function. Then $x \in \mathrm{Det}(b)$ if and only if there is $c \in (\alpha, t(x))$ with $J^-(x) \cap S_c \subseteq I^-_{\mathrm{s}}(b)$.*

Proof. ($\Leftarrow$) A past-inextendible causal curve from x crosses the Cauchy surface S_c, at a point of $J^-(x) \cap S_c \subseteq I^-_{\mathrm{s}}(b)$.

($\Rightarrow$) Suppose that for every $c \in (\alpha, t(x))$ the set $K_c = (J^-(x) \cap S_c) \setminus I^-_{\mathrm{s}}(b)$ is nonempty; it is compact, since $I^-_{\mathrm{s}}(b)$ is open. Fix $t(x) > c_1 > c_2 > \dots$ with $c_n \to \alpha$, and let

$$P_n = \{(z_1, \dots, z_n) \in K_{c_1} \times \cdots \times K_{c_n} : x \ge z_1 \ge z_2 \ge \cdots \ge z_n\}.$$

P_n is compact, since $\le$ is closed. It is nonempty: pick $y \in K_{c_n}$ and a causal curve from x to y; it crosses each S_{c_i}, $i < n$, at some $z_i \le x$, and $z_i \notin I^-_{\mathrm{s}}(b)$, for otherwise $y \le z_i \ll b$ would give $y \ll b$. The maps $P_{n+1} \to P_n$ forgetting the last coordinate are continuous, so the inverse limit of the nonempty compact spaces P_n is nonempty, giving $x \ge z_1 \ge z_2 \ge \cdots$ with $z_n \in K_{c_n}$. Join consecutive terms by causal curves and concatenate. Each segment avoids $I^-_{\mathrm{s}}(b)$ by push-up, as before. Along the concatenation $t \to \alpha$, so it has no past endpoint in M: it is a past-inextendible causal curve from x missing $I^-_{\mathrm{s}}(b)$. □

Corollary 4. *If $I^-_{\mathrm{s}}(b)$ contains a Cauchy surface S, then $\mathrm{Det}(b) = M$.*

Proof. If $x \in I^-_{\mathrm{s}}(S) \cup S$, then $x \le s \ll b$ for some $s \in S$, so $x \in I^-_{\mathrm{s}}(b)$. If $x \in I^+_{\mathrm{s}}(S)$, extend a past-inextendible causal curve from x to an inextendible one; it meets S, and not to the future of x, since then $s' \ll x \le s$ for some $s, s' \in S$, contradicting achronality. So the curve from x meets $S \subseteq I^-_{\mathrm{s}}(b)$. □

Since $\mathrm{Det}(b)$ is defined by causal curves alone, it depends only on the conformal class of the metric; conformal diagrams decide it.

3 The standard models

Lemma 5. *Let (Σ, d) be $\mathbb{R}^3$, hyperbolic space H^3, or the unit sphere S^3 with $r_2 \le \pi$. Then the closed ball $\bar{B}(X, r_1)$ lies in the open ball $B(Y, r_2)$ if and only if $d(X, Y) + r_1 < r_2$.*

Proof. ($\Leftarrow$) is the triangle inequality. ($\Rightarrow$) Suppose $d(X, Y) + r_1 \ge r_2$. If $d(X, Y) + r_1 \le \pi$ (always so on $\mathbb{R}^3$ and H^3), follow a minimizing geodesic from Y through X (any geodesic from X if $X = Y$) a further distance r_1; the endpoint lies in $\bar{B}(X, r_1)$ at distance $d(X, Y) + r_1 \ge r_2$ from Y. If $d(X, Y) + r_1 > \pi$ on S^3, the antipode of Y is at distance $\pi - d(X, Y) < r_1$ from X and $\pi \ge r_2$ from Y. □

Proposition 6. *Let M be conformal to $(\eta_-, \eta_+) \times \Sigma$ with metric $-d\eta^2 + d\Sigma^2$, where Σ is $\mathbb{R}^3$, H^3, or the unit S^3 with $\eta_+ - \eta_- \le \pi$. Then $\mathrm{Det}(b) = I^-_{\mathrm{s}}(b)$ for every b.*

Proof. Use $t = \eta$. For $x = (\eta_x, X)$, $b = (\eta_b, Y)$: $x \ll b$ iff $d(X, Y) < \eta_b - \eta_x$ (on S^3, $\eta_b - \eta_x < \pi$). For $\eta_- < c < \eta_x$, $J^-(x) \cap S_c = \bar{B}(X, \eta_x - c)$ and $I^-_{\mathrm{s}}(b) \cap S_c = B(Y, \eta_b - c)$, with $\eta_b - c < \pi$ on S^3. By Lemma 5 the inclusion holds iff $d(X, Y) < \eta_b - \eta_x$, i.e. iff $x \ll b$. Apply Theorem 3. □

This covers Minkowski spacetime, spatially flat and open FLRW universes with a big bang (conformal to $\eta > 0$), the closed radiation universe ($a \propto \sin\eta$, $\eta \in (0, \pi)$) and global de Sitter spacetime ($\Sigma = S^3$, $\eta \in (-\pi/2, \pi/2)$) [12]. In none of them is anything outside an event's chronological past determined from it.

Proposition 7. *Let M be conformal to $(\eta_-, \eta_+) \times S^3$ with $\eta_+ - \eta_- > \pi$. Then $\mathrm{Det}(b) = I^-_{\mathrm{s}}(b)$ if $\eta_b \le \eta_- + \pi$, and $\mathrm{Det}(b) = M$ if $\eta_b > \eta_- + \pi$. In the Einstein static universe ($\eta_\pm = \pm\infty$), $\mathrm{Det}(b) = M$ for every b.*

Proof. If $\eta_b > \eta_- + \pi$, take $c \in (\eta_-, \eta_b - \pi)$: $I^-_{\mathrm{s}}(b) \cap S_c$ is an open ball of radius greater than π, hence all of S_c, and Corollary 4 applies. If $\eta_b \le \eta_- + \pi$, every $c > \eta_-$ has $\eta_b - c < \pi$, and the proof of Proposition 6 applies. □

For the closed dust FLRW model, $a(\eta) = \frac{1}{2} a_m (1 - \cos\eta)$ with $\eta \in (0, 2\pi)$, the threshold is $\eta = \pi$, the moment of maximum expansion. That the particle horizon of the closed dust model disappears at maximum expansion, so that an observer can then in principle see the whole universe, is known [3, 10]; the Einstein static case is Geroch's [8, Fig. 5]. What Proposition 7 adds is the consequence for determination: with Corollary 4, the vanishing of the particle horizon is exactly the condition under which what is real from b fixes everything. The result fits Manchak's theorem that prediction of events to the future of an observer requires a compact Cauchy surface [14]: only spatially closed models can cross the threshold, and not all of them do.

4 Unreality and openness

Put together with the companion theorem, these results separate two properties of an event x relative to a standpoint b:

- *real from* b: x is in b's fragment, which a local becoming theory fixes to lie between $I^-_{\mathrm{s}}(b)$ and $J^-(b)$;
- *settled from* b: under laws with the domain-of-dependence property, $x \in \mathrm{Det}(b)$.

Every real event (with the possible exception of the null boundary) is settled. The converse fails in two ways. In the flat, open, radiation and de Sitter models nothing beyond the fragment is settled; in the closed dust model after maximum expansion, and in the Einstein static universe, everything is. So:

1. *Unreality does not entail openness.* An event can be unreal from b and settled from b: in the closed dust model, every future event is so from every event after maximum expansion. This is the relativistic form of the determinist growing-block position [4], and it is also where Rakíc's intuition belongs [16]: her sea-fight is, at most, settled from the observer, not real from her. In Minkowski spacetime it is not even settled: by Proposition 6, nothing outside the observer's past is, whatever lies in that past.
2. *Determinism does not entail settledness from each event.* Under deterministic laws the whole spacetime is fixed by data on any Cauchy surface, but no single event's past contains a Cauchy surface in the flat, open, radiation or de Sitter models. From each standpoint, the future is then unsettled as well as unreal.
3. *Which combination holds is a matter of global causal structure and can change with cosmic epoch.* Given the laws, the answer to "is the future open from here?" can differ between two events of the same spacetime.

Three limits. First, this openness is relational and informational: an event outside $\mathrm{Det}(b)$ is still fixed by its own past, and a determinist eternalist can accept everything in this section. What the section shows is that the standpoint theorist's claim that the future is unreal from b neither secures nor excludes its being fixed from b; an ontically open future needs indeterministic laws, about which $\mathrm{Det}(b)$ is silent. Second, the closed dust result concerns an idealized model with deterministic, classical dynamics; it is not a claim about our universe, which on current evidence is not of that kind. Third, Manchak shows that an observer who does not already know the spacetime cannot verify such predictions [14]; $\mathrm{Det}(b)$ is a nomological notion given M, not an epistemic one.

5 A remark on standpoint logic

Read $\Box A$ at b as "at every event $x \leq b$, A" on the frame (M, R) with bRx iff $x \leq b$. If A is a tenseless claim about what occurs at x, this is the reflexive past modality of Diodorean tense logic, and the standpoint reading of Fine and Lipman [7, 13] adds no formal content. R is reflexive and transitive, so S4 is valid, and in Minkowski spacetime the valid formulas are exactly those of S4.2 [9]. On reflexive transitive frames the axiom .2, $\Diamond\Box A \to \Box\Diamond A$, is valid iff the frame is directed [1]; on the subframe generated by b, directedness says that any two events in $J^-(b)$ have a common causal past.

In a model conformal to $(\eta_-, \eta_+) \times \mathbb{R}^3$, two events have a common past iff $|X-Y| < \eta_x + \eta_y - 2\eta_-$, so if $\eta_- > -\infty$ the directedness fails in the past of every event and .2 is not valid on any generated subframe. This is the particle-horizon structure of Rindler [17] in modal form, and it holds equally with or without an inflationary era of finite conformal duration. The horizon problem of cosmology [11] is therefore not the failure of .2, which holds on both sides of it, but the failure of particular instances: pairs of points on our last-scattering surface, at comoving separation up to $2(\eta_0 - \eta_{\mathrm{ls}})$, have no common past unless $\eta_0 - \eta_{\mathrm{ls}} < \eta_{\mathrm{ls}} - \eta_-$, which fails without inflation and holds with enough of it. Two cautions. Under the view's persistence principle, on which what holds from an event holds from every later one, valuations are constrained along the future direction, and the frame property that matters may concern common futures, which these models have; the logic of persistent valuations is a separate question. And Lewis and van der Schaaf study causal modal logics of general spacetimes in the future direction [21]; the past-directed frames used here can behave differently. Whether S4 is exactly the logic of all past-directed globally hyperbolic frames we leave open.

6 Conclusion

The unreality of the future from an event, which local theories of relativistic becoming deliver, neither implies nor excludes its being fixed from that event. Which holds is decided, given deterministic local laws, by a slicing condition on Geroch's domain of prediction (Theorem 3), and it is computable from conformal diagrams. In most standard models nothing beyond an event's past is fixed from it; in the closed dust model everything is, from every event after maximum expansion. The question whether the future is open from a standpoint thus divides into a question about the laws and a question about global causal structure, and only the second is settled by relativity.

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